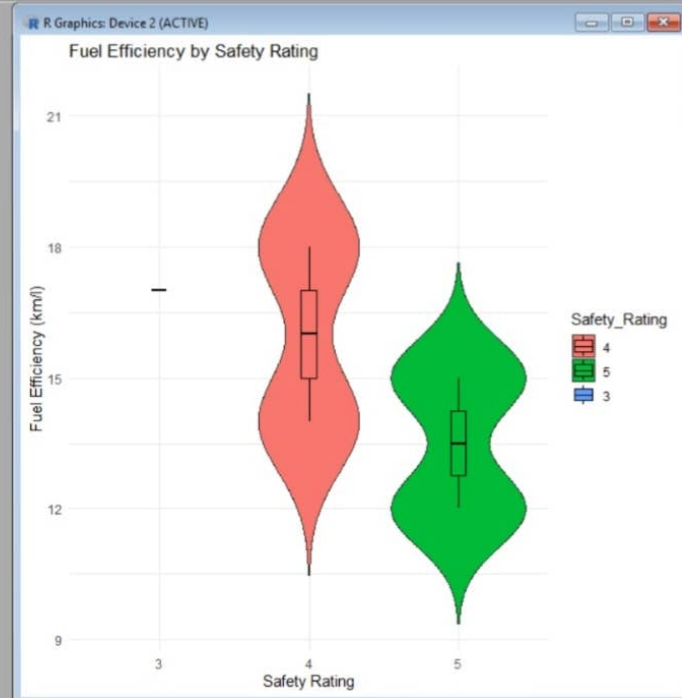


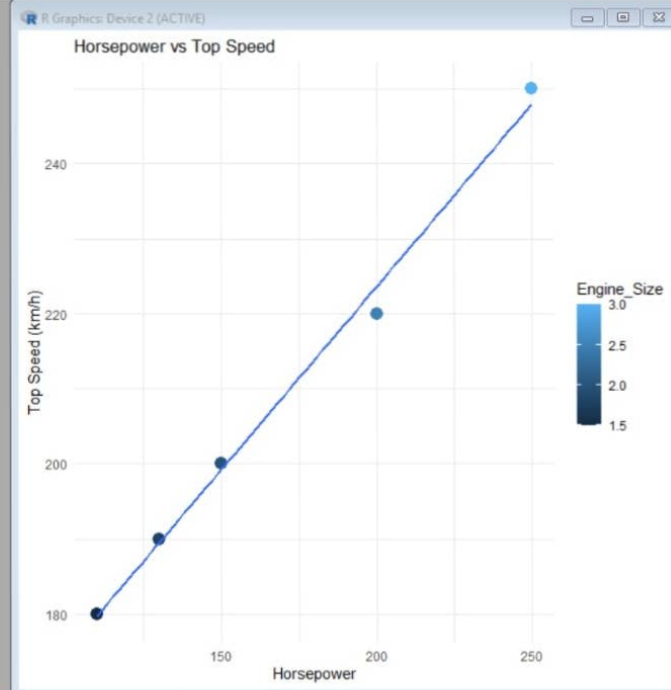
```
> library(ggplot2)
>
> # Dataset
> vehicle <- data.frame(
+   Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
+   Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
+   Horsepower = c(110, 150, 250, 200, 130),
+   Fuel_Efficiency = c(18, 15, 12, 14, 17),
+   Top_Speed = c(180, 200, 250, 220, 190),
+   Safety_Rating = as.factor(c(4, 5, 5, 4, 3))
+ )
>
> # Violin Plot
> ggplot(vehicle, aes(x = Safety_Rating, y = Fuel_Efficiency, fill = Safety_Rating)) +
+   geom_violin(trim = FALSE) +
+   geom_boxplot(width = 0.1, color = "black") +
+   labs(title = "Fuel Efficiency by Safety Rating",
+        x = "Safety Rating",
+        y = "Fuel Efficiency (km/l)" +
+   theme_minimal()
Warning message:
Groups with fewer than two datapoints have been dropped.
[1] Set 'drop = FALSE' to consider such groups for position adjustment purposes.
>
```



```

+ geom_violin(trim = FALSE) +
+ geom_boxplot(width = 0.1, color = "black") +
+ labs(title = "Fuel Efficiency by Safety Rating",
+       x = "Safety Rating",
+       y = "Fuel Efficiency (km/l)") +
+ theme_minimal()
Warning message:
Groups with fewer than two datapoints have been dropped.
[1] Set 'drop = FALSE' to consider such groups for position adjustment purposes.
> ggplot(vehicle, aes(x = Horsepower, y = Top_Speed, color = Engine_Size)) +
+   geom_point(size = 4) +
+   geom_smooth(method = "lm", se = FALSE) +
+   labs(title = "Horsepower vs Top Speed",
+         x = "Horsepower",
+         y = "Top Speed (km/h)") +
+   theme_minimal()
'geom_smooth()' using formula = 'y ~ x'
Warning message:
The following aesthetics were dropped during statistical transformation:
colour.
[1] This can happen when ggplot fails to infer the correct grouping structure in
the data.
[1] Did you forget to specify a 'group' aesthetic or to convert a numerical
variable into a factor?
>
4

```



```

Installing package into 'C:/Users/sandr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://cran.isid.ac.in/bin/windows/contrib/4.6/corrplot_0.95.zip'
Content type 'application/zip' length 3827329 bytes (3.7 MB)
downloaded 3.7 MB

package 'corrplot' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:\Users\sandr\AppData\Local\Temp\RtmpEFxMCO\downloaded_packages
> library(corrplot)
corrplot 0.95 loaded
> library(corrplot)
>
> # Convert dataset (remove categorical)
> num_data <- vehicle[, c("Engine_Size", "Horsepower", "Fuel_Efficiency", "Top_Speed")]
>
> # Correlation matrix
> corr_matrix <- cor(num_data)
>
> # Heatmap
> corrplot(corr_matrix, method = "color", addCoef.col = "black",
+         tl.col = "black", number.cex = 0.8)
>

```

